Responses to Selection Criteria

Wentao Zhang · Application for Lecturer / Senior Lecturer in Electrical Machines · SAE-031274
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The responses below address each item of the **Person Specification** for the Lecturer / Senior Lecturer in Electrical Machines post (SAE-031274). The seven Essential criteria are answered first, followed by the three Strongly Desirable criteria. Cited publications are listed at the end of the document.

1. PhD in Electrical Engineering (Essential)

I hold a **PhD in Electrical Engineering** from Southeast University (SEU), Nanjing, China (Sep 2021 to Sep 2025), with the dissertation *Multi-physics Analysis and Design of Hairpin Winding PM Traction Machines in Electric Vehicles*. My MSc (2018 to 2021) and BEng (2014 to 2018) are also in Electrical Engineering from SEU. My doctoral research, supervised by Prof. Zhongze Wu, focused on the **electromagnetic design, advanced thermal modelling, and control of high power-density electrical machines**, which align directly with the research focus of this post. The work was recognised by the **2025 IEEE Trans. Transportation Electrification Prize Paper Award (2nd Prize)** and the **2025 IEEE TEC Electric Vehicle Design Competition (1st Prize)**.

2. Industrial Experience including FEA of Electromagnetic Circuits, Machine Design, Machine Prototyping Methods and Laboratory Testing (Essential)

During my PhD I collaborated with **Inovance**, a world-leading EV traction-machine supplier, on the analysis, design, prototyping, and laboratory testing of two advanced high power-density oil-cooled PM traction machines:

- **18 000 rpm PM traction machine for A0-class vehicle**: measured peak power 113 kW, peak torque 138 Nm, power density 7.78 kW/kg, peak efficiency 95.3%.
- **20 000 rpm PM traction machine for A-class vehicle**: measured peak power 270 kW, peak torque 380 Nm, power density 7.87 kW/kg, peak efficiency 97.4%.

Both projects followed full industrial development cycles. At the design stage, I applied a **multi-physics optimisation framework** coupling electromagnetic field analysis, thermal modelling, and mechanical stress evaluation. At the manufacturing stage, I produced engineering drawings for the electromagnetic and structural components, gaining substantial experience in tolerance specification and high-reliability manufacturing. At the testing stage, I characterised both prototypes on an advanced **AVL dynamometer system**, extracting key performance indicators, voltage and current waveforms, and full efficiency maps.

In addition, I have developed an **in-house finite element analysis code package** covering electromagnetic, structural-modal, and mechanical-stress analyses of electrical machines. This package is now being applied to magnet-free synchronous machine development in my current Research Assistant role at PEMC, University of Nottingham.

3. Outstanding Track Record of Publications (Essential)

I have authored **22 journal papers, including 17 in IEEE Transactions**, across *IEEE Trans. Industrial Electronics*, *Industry Applications*, *Transportation Electrification*, and *Energy Conversion*. I have also presented **18 conference papers** at leading international venues including IEEE IEMDC, IEEE ECCE, ICEMS, and ICEM, and hold **9 patents**.

The research themes of my publications align directly with the Department's electrical-machines portfolio:

- **Novel machine design and winding technology** — novel *n*-layer flat-wire / hairpin winding design methodology for high power-density EV PM machines, including conductor layout optimisation, mechanical design, and experimental validation.
- **Electromagnetic modelling and analysis** — systematic analytical and FEA-based investigation of air-gap field modulation, DC winding-induced voltage, end-effects, and flux regulation in wound-field switched flux machines and hybrid excitation topologies.
- **Advanced thermal management** — high-fidelity lumped-parameter thermal network modelling and fluid-solid coupled oil-cooling analysis for hairpin-winding PM machines, validated against prototype measurements.

- **Multi-physics optimisation** — mechanism-driven multi-objective optimisation integrating electromagnetic performance, thermal behaviour, and mechanical constraints across multiple machine topologies.
- **Machine control and dynamic modelling** — spatial harmonic modelling of machine inductance, DC winding current dynamics under PWM excitation, and converter-machine interaction analysis for wound-field and PM machine systems.

4. Recognition in the International Research Community (Essential)
My work has received competitive recognition at journal and conference level:

- **2025 1st Prize, IEEE TEC Electric Vehicle Design Competition**
- **2025 2nd Prize Paper Award, IEEE Trans. Transportation Electrification**
- 2024 2nd Prize, 11th Energy Equipment Innovative Design Competition for China Postgraduates
- 2023 Best Paper Award, China International Electrical Energy Conference (CIEEC)
- **2023 1st Prize, IEEE IAS CMD Thesis Contest (non-PhD category)**
- 2022 Outstanding Master Degree Thesis Award, Jiangsu Province

I have presented work at leading international conferences including IEEE ECCE, IEEE IEMDC, ICEM, and ICEMS (in person and through remote sessions during the COVID-era programmes), engaging directly with the international electrical-machines research community on novel machine design, multi-physics analysis, and advanced winding technology. I also serve as **Co-Organiser and Co-Chair of an ICEM 2026 Special Session** (Funchal, Madeira, Portugal) on *Next-Generation Electrified Transport: Advances in Electric Drive Systems*, jointly with Dr Lingyun Shao (NUAA).

5. Potential for Attracting External Research Funding (Essential)
My research programme addresses priorities that are central to major UK and international funding bodies. The electrification of transport, including advanced machine design for EVs and aircraft, is a strategic focus of **EPSRC**, **Innovate UK**, and the **Aerospace Technology Institute (ATI)**. My multi-physics approach to traction-machine design, with experimental validation on dynamometer-tested prototypes, is well aligned with current EPSRC priority areas in transport decarbonisation and engineering for net zero, and with ATI's roadmap on more-electric and hybrid-electric propulsion.

Beyond academic funding, my industry-standard design and testing experience through the **Inovance** collaboration positions me to bridge academic research and commercial application, supporting **direct industrial funding** and **co-funded R&D collaborations** with leading EV and aerospace machine manufacturers. My current postdoctoral programme at PEMC Nottingham, on next-generation magnet-free synchronous machines, also provides a natural pathway to multi-institution UK-funded consortia in transport electrification.

6. Oral and Written Communication Skills (Essential)
Written communication is evidenced by **22 journal papers and 18 conference papers** in leading English-language IEEE journals and proceedings, including a **prize paper award** at *IEEE Trans. Transportation Electrification* where clarity and presentation quality form part of the assessment. Oral communication is evidenced by presentations at major international conferences including IEEE ECCE, IEEE IEMDC, ICEM, and ICEMS, departmental seminars at SEU and Nottingham, and my upcoming role chairing an ICEM 2026 Special Session before an international audience.

In my current role at PEMC Nottingham, I regularly engage with academic colleagues and industrial partners through technical project meetings, research seminars, and progress reviews. I also serve as a peer reviewer for several IEEE journals, including *IEEE Trans. Industrial Electronics*, *Industry Applications*, *Energy Conversion*, and *Transportation Electrification*.

7. Ability to Deliver Teaching in Electrical Machines and Drives (Essential)
My doctoral and postdoctoral research provides direct expertise in the subjects that form the core of undergraduate and postgraduate **Electrical Machines and Drives**, covering electromagnetic machine theory, principles of finite element analysis, FEA-based analysis and design of electrical machines, power-converter interactions, thermal management, and machine control. I am happy to contribute to existing modules at undergraduate and MSc level, and to develop new teaching content reflecting contemporary developments in traction-machine technology of high relevance to students pursuing careers in EV and aerospace electrification, including **novel cooling methods** and **hairpin-winding layouts**.

During my PhD I served as **teaching assistant** for the SEU postgraduate course *Integrated Practice of Electrical Machine Systems*. My responsibilities covered preparing and delivering tutorial slides for selected sessions, running daily Q&A and consultation hours, reviewing student final-project outline proposals with structured written feedback on scope and methodology, and providing mark recommendations to the course coordinator. This gave me direct exposure to the postgraduate teaching cycle.

To support practical and project-based teaching, my toolchain spans MATLAB/Simulink, ANSYS Maxwell and Mechanical, Python, C/C++ for TI C2000 and STM32, and FPGA, complemented by hands-on power-electronics prototyping (a three-phase full-bridge converter and its control algorithm, built and commissioned for testing flux-modulation machine prototypes) and an in-house Rust MCU data-acquisition tool, both of which can be adapted into laboratory and capstone material.

The following responses address the three **Strongly Desirable** criteria.

(a) Cryogenic Systems
While cryogenic machine design has not been the primary focus of my research, my **advanced thermal-modelling expertise** provides a rigorous foundation for engaging with cryogenic cooling challenges. In particular, I have developed **high-fidelity lumped-parameter thermal networks** and **coupled fluid-thermal analyses** for hairpin-winding PM machines, validated against prototype dynamometer measurements. These modelling techniques are directly transferable to cryogenically cooled machines, where accurate prediction of local hot-spots, conductor temperature gradients, and coolant flow distribution is similarly critical. I am genuinely enthusiastic about developing research capability in this area alongside Manchester's established strengths in superconducting and HTS-machine research, and would welcome the opportunity to contribute to existing or new project proposals in cryogenic and superconducting machines.

(b) Health and Safety in University Laboratories
Throughout my doctoral study at SEU and my Research Assistant role at PEMC Nottingham, I have worked in **high-voltage electrical-machine test facilities** operating within **formal health and safety frameworks**, including documented risk assessments, laboratory induction procedures, and safe working procedures for rotating machinery and power-electronics test rigs. I have personally followed lock-out / tag-out protocols on dynamometer test cells, observed permit-to-work procedures during high-voltage battery-emulator operation, and supervised junior students performing measurement campaigns. I appreciate fully the responsibility of managing laboratory environments and equipment for student use, and would commit to completing Manchester's required laboratory-supervisor and risk-assessment training in my first appointment year.

(c) Power Electronics and AC Power Circuit Analysis
A substantial strand of my research addresses the interaction between power-electronic drives and electrical machines, including the **effects of PWM excitation on machine losses and temperature**, **field-winding voltage modulation methods**, and **induced voltage pulsation**. Beyond modelling, I have designed and implemented a **three-phase full-bridge converter and associated control algorithm** for testing flux-modulation machine prototypes, covering hardware design, gate-drive selection, sensor integration, control-software development on a TI C2000 platform, and laboratory commissioning. This combination of analytical, simulation, and hardware-implementation experience provides a strong basis for contributing to teaching in **power electronics and AC power-circuit analysis**, and I would welcome the opportunity to complement the Department's existing curriculum in these areas.

The complete list of journal papers, conference papers, and patents referenced above is provided in the attached CV.